

Yash Baruah

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EDUCATION

University of Texas at Dallas

Bachelor of Science in Computer Science

Richardson, TX

Expected Dec. 2026

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Advanced Algorithm Design, Database Systems, Software Engineering, Systems Programming, Probability & Statistics.

EXPERIENCE

Nokia

AI Software Automation Co-op

Dallas, TX

May 2026 – Aug. 2026

- Evaluated the **LLM assistant** in Nokia's optical network-management platform across **~20** use cases (live **alarms**, **topology**, inventory, and API guidance), converting findings into **6** defect tickets and **10** roadmap proposals for R&D.
- Ran **adversarial safety probes** (credential and customer-data extraction attempts, fabricated-feature traps), verifying the assistant's honest refusals, then designed **regression tests** to protect those behaviors through an LLM upgrade.
- Exposed an AI assistant silently **overriding explicit user parameters** (a noGMPLS design returned **GMPLS-enabled** into a real bill of materials) by executing a **38-prompt** plan with paired chat-vs-UI evidence per test.
- Authored a **61-test**, 10-phase regression kit tracing every baseline defect to re-test IDs, with indirect **prompt-injection** coverage and **6 fault-injection** scripts designed around known faults for **ground-truth** scoring.

Mashfrog

Software Engineering Intern

Remote

May 2025 – Aug. 2025

- Developed and validated an automation bot, achieving **98%** accuracy across **2,000+** test cases while reducing manual errors and strengthening exception handling.
- Designed, tested, and deployed **20+ RPA** and **generative AI** workflows that automated core business processes, saving over **30 hours** of manual effort weekly and accelerating task completion by up to **400%**.

PROJECTS AND RESEARCH

HaltIQ | ACM Research, UT Dallas

Fall 2026

- Selected for a **four-student research team** to investigate reinforcement learning for cost-efficient delegation in multi-agent AI systems.
- Research combines **GRPO**, potential-based reward shaping, and selective **LLM-as-judge** evaluation to study cost, latency, and quality trade-offs.

ToyotaTinder: AI Car Matchmaker | Next.js, React, Node.js, Google Gemini

Nov. 2025

- Built a **Next.js** App Router web app for **HackUTD 2025** pairing **Toyota inventory data** with **Google Gemini 2.5 Flash** to generate personalized car recommendations.
- Implemented server-side filtering, scoring, robust **JSON parsing**, and fallback handling to keep AI recommendations reliable under malformed model output.

VisionTranslate | Chrome MV3, React, JavaScript, OCR, LLMs

Apr. 2026 – Present

- Developed a **Chrome MV3** extension that detects images, canvases, and background images on any webpage, runs **OCR**, and renders layout-aware translated overlays in place.
- Engineered the end-to-end **OCR-to-LLM** translation pipeline with caching for low-latency overlays.

AI Music Genre Classification | Python, TensorFlow, Librosa, AWS Lambda

UTD AIS

- Trained a **TensorFlow** CNN on audio spectrograms with **Librosa** preprocessing, normalization, and augmentation, reaching **~90%** genre-classification accuracy with TensorBoard experiment tracking.
- Served real-time genre inference through **AWS Lambda** behind a React front end.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, JavaScript, TypeScript, SQL
- **AI & Security Testing:** Adversarial & Safety Testing, Prompt-Injection Test Design, Fault-Injection (Known-Answer) Test Design, LLM/RAG Evaluation, Baseline Regression Scoring, Test Coverage & Traceability
- **Networking:** REST APIs, WebSockets, Fault & Alarm Analysis, Optical Transport Exposure (DWDM, OTN)
- **Tools & Cloud:** Linux/UNIX, Git, Node.js, FastAPI, MongoDB, RPA, AWS Lambda, AWS S3
- **Certifications:** AWS Certified Cloud Practitioner, Databricks Fundamentals Accreditation